

Part X

Linear Regression

Lecture Outline

① Linear Regression

② Regularization in Linear Regression

Linear Regression Form

- Consider that we have N input-output pairs $\{(x_1, y_1), \dots, (x_N, y_N)\}$ with $x_n \in \mathbb{R}^d, y_n \in \mathbb{R}, n \in \{1, \dots, N\}$.
- Example: CO2 prediction: x is time and y is CO2 concentration in ppm.

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- The goal is to fit a model $g(x)$ such that for a new test point x_{test} , we can predict the output $\hat{y}_{\text{test}} = g(x_{\text{test}})$.

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- Linear regression assumes the following parametric model

$$g(x) = \sum_{i=0}^{M-1} w_i \phi_i(x),$$

where ϕ_i here are **chosen** *basis* functions.

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where ϕ_i here are **chosen** *basis* functions.

- Examples:
 - A line: $\phi_0(x) = 1, \phi_1(x) = x_1, \phi_2(x) = x_2$.
 - Fourier basis: $\phi_1 = 1, \phi_2 = \sin(2\pi f x_1), \phi_3 = \sin(2\pi f x_2)$, etc...
 - Gaussian kernels.
 - Polynomials.

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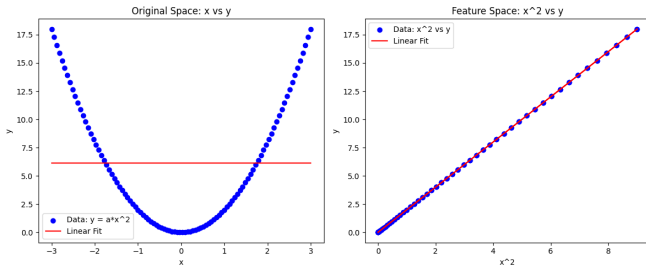
- What did we choose for the basis of the CO2 exercise?

Linearity in the feature space

- The basis functions $\phi_0(x), \dots, \phi_{M-1}(x)$ should be selected such that the data is linear in the *feature space*.

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- Example:



Recap: Constant+Gaussian

Fact: if we have a constant $c \in \mathbb{R}$ and a gaussian random variable $E \sim \mathcal{N}(\mu, \sigma^2)$

($p_E(e) = \frac{1}{2\sqrt{\pi\sigma^2}} \exp(\frac{-(e-\mu)^2}{2\sigma^2})$), then $Y = c + E \sim \mathcal{N}(\mu + c, \sigma^2)$

($p_Y(y) = \frac{1}{2\sqrt{\pi\sigma^2}} \exp(\frac{-(y-\mu-c)^2}{2\sigma^2})$) .

Least-Squares and Maximum Likelihood

- We assume that our model for the linear regression is

$$\hat{y}_n = g(x_n) + e_n = \sum_{i=0}^{M-1} w_m \phi_m(x_n) + e_n = w^\top \phi(x_n) + e_n, \quad n \in \{1, \dots, N\},$$

where $e_1, \dots, e_n \sim \mathcal{N}(0, 1)$ are i.i.d. measurement noise,
 $w = [w_0, \dots, w_{M-1}]^\top$, $\phi(x_n) = [\phi_0(x_n), \dots, \phi_{M-1}(x_n)]^\top$.

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- In vector form

$$\hat{\mathbf{y}} = \boldsymbol{\Phi}(\mathbf{x})\mathbf{w} + \mathbf{e},$$

where $\hat{\mathbf{y}} = [\hat{y}_1, \dots, \hat{y}_N]^\top$, $\mathbf{x} = [x_1, \dots, x_N]^\top$, $\mathbf{e} = [e_1, \dots, e_N]^\top \sim \mathcal{N}(0, \sigma^2 \mathbf{I})$, and

$$\boldsymbol{\Phi}(\mathbf{x}) = \begin{pmatrix} \phi_0(x_1) & \phi_1(x_1) & \cdots & \phi_M(x_1) \\ \phi_0(x_2) & \phi_1(x_2) & \cdots & \phi_M(x_2) \\ \vdots & \vdots & \ddots & \vdots \\ \phi_0(x_N) & \phi_1(x_N) & \cdots & \phi_M(x_N) \end{pmatrix}.$$

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- In vector form

$$\hat{\mathbf{y}} = \Phi(\mathbf{x})\mathbf{w} + \mathbf{e},$$

- We want to obtain the MLE $\hat{\mathbf{w}}_{\text{MLE}}$.

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- In vector form

$$\hat{\mathbf{y}} = \boldsymbol{\Phi}(\mathbf{x})\mathbf{w} + \mathbf{e},$$

- We want to obtain the MLE $\hat{\mathbf{w}}_{\text{MLE}}$.
- The likelihood is

$$\begin{aligned} p_{\mathbf{Y}|\mathbf{X}, \mathbf{W}}(\mathbf{y} \mid \mathbf{x}, \mathbf{w}) &= \prod_{n=1}^N p_{Y|X, W}(y_n \mid x_n, w) \\ &= \prod_{n=1}^N \frac{1}{2\sqrt{\pi}\sigma^2} \exp\left(-\frac{\left(y_n - \sum_{m=0}^{M-1} w_m \phi_m(x_n)\right)^2}{2\sigma^2}\right) \end{aligned} \quad (1)$$

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- In vector form

$$\hat{\mathbf{y}} = \boldsymbol{\Phi}(\mathbf{x})\mathbf{w} + \mathbf{e},$$

- We want to obtain the MLE $\hat{\mathbf{w}}_{\text{MLE}}$.
- The log-likelihood is then

$$\begin{aligned} \ell(\mathbf{w}) = \ln(p_{\mathbf{Y}|\mathbf{X}, \mathbf{W}}(\mathbf{y} | \mathbf{x}, \mathbf{w})) &= \frac{1}{2\sigma^2} \sum_{n=1}^N - \left(y_n - \sum_{m=0}^{M-1} w_m \phi_m(x_n) \right)^2 - \frac{1}{2} \ln(2\pi\sigma^2) \\ &= \frac{-1}{2\sigma^2} \|\mathbf{y} - \boldsymbol{\Phi}(\mathbf{x})\mathbf{w}\|^2 - \frac{1}{2} \ln(2\pi\sigma^2) \end{aligned}$$

- The MLE estimate is

$$\hat{\mathbf{w}}_{\text{MLE}} = \left(\boldsymbol{\Phi}^\top(\mathbf{x})\boldsymbol{\Phi}(\mathbf{x}) \right)^{-1} \boldsymbol{\Phi}^\top(\mathbf{x})\mathbf{y}$$

It is the *same solution* as the Least-Squares solution!

Multiple Outputs

- So far, we considered the case of one output $y \in \mathbb{R}$. We can generalize it to be K $y \in \mathbb{R}^K$ outputs are write

$$\hat{y}_n = \mathbf{w}^\top \phi(x_n) + e_n, \quad n \in \{1, \dots, N\},$$

where $\mathbf{w} \in \mathbb{R}^{M \times K}$ is the weight (coefficients) *matrix*, and $e_n \sim \mathcal{N}(0, \sigma^2 \mathbf{I}_{K \times K})$.

- The matrix form for all the data

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where $\hat{\mathbf{y}} = \begin{bmatrix} \hat{y}_1^\top \\ \vdots \\ \hat{y}_n^\top \end{bmatrix} \in \mathbb{R}^{N \times K}$, $\mathbf{e} = \begin{bmatrix} e_1^\top \\ \vdots \\ e_n^\top \end{bmatrix} \in \mathbb{R}^{N \times K}$, and

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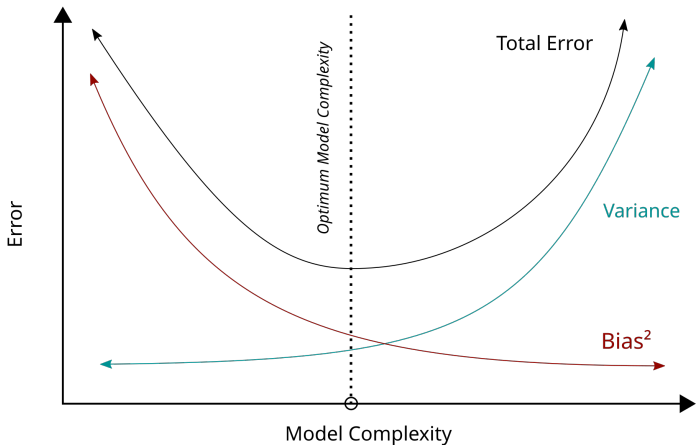
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- The MLE estimate is

$$\hat{\mathbf{w}}_{\text{MLE}} = (\Phi^\top(\mathbf{x})\Phi(\mathbf{x}))^{-1}\Phi^\top(\mathbf{x})\mathbf{y}$$

Recap: Bias-variance tradeoff

- There is always a trade-off between bias (underfitting) and variance (overfitting), with very flexible models having low bias and high variance, and relatively rigid models having high bias and low variance.
- This problem has not been solved completely yet!



Recap: Bias-variance tradeoff

- There is always a trade-off between bias (underfitting) and variance (overfitting), with very flexible models having low bias and high variance, and relatively rigid models having high bias and low variance.
- This problem has not been solved completely yet!
- One way to control the model's flexibility is using *regularization*.

Need for Regularization

- Problem: Least squares solutions can lead to *overfitting* when:
 - Number of basis functions M is large (high complexity)
 - Training data is limited (N small)

Need for Regularization

- Problem: Least squares solutions can lead to *overfitting* when:
 - Number of basis functions M is large (high complexity)
 - Training data is limited (N small)
- Solution: Add *regularization term* to objective function

$$J(w) = \underbrace{\frac{1}{2} \|\mathbf{y} - \Phi(\mathbf{x})w\|^2}_{\text{Data fit}} + \underbrace{\lambda R(w)}_{\text{Regularizer}}$$

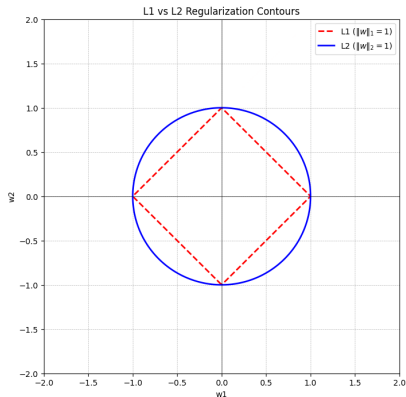
where $\lambda > 0$ controls regularization strength

Need for Regularization

- Problem: Least squares solutions can lead to *overfitting* when:
 - Number of basis functions M is large (high complexity)
 - Training data is limited (N small)
- Common regularization types:
 - L2 (Ridge): $R(w) = \frac{1}{2} \|w\|_2^2 = \frac{1}{2} \sum_{i=0}^{M-1} w_m^2$
 - L1 (Lasso): $R(w) = \|w\|_1 = \sum_{i=0}^{M-1} |w_m|$
 - Elastic Net: Combination of L1 + L2.

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MAP Interpretation of Regularization (1D output)

- Recall Bayesian approach: Maximum A Posteriori (MAP) estimation

$$\hat{w}_{\text{MAP}} = \operatorname{argmax}_{w \in \mathbb{R}^M} p_{W|\mathbf{Y}, \mathbf{X}}(w|\mathbf{y}, \mathbf{x}) = \operatorname{argmax}_{w \in \mathbb{R}^M} p_{\mathbf{Y}|\mathbf{X}, W}(\mathbf{y}|\mathbf{x}, w) p_W(w)$$

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- For Ridge Regression (L2):
 - Likelihood: $p_{\mathbf{Y}|\mathbf{X}, W}(\mathbf{y}|\mathbf{x}, w) = \mathcal{N}(\Phi(\mathbf{x})w, \sigma^2 I)$
 - Prior: $p_W(w) = \mathcal{N}(0, \tau^2 I)$

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- Taking negative log-posterior:

$$-\ln p_{W|Y, X}(w|y, x) = \frac{1}{2\sigma^2} \|y - \Phi(x)w\|_2^2 + \frac{1}{2\tau^2} \|w\|_2^2 + \text{const}$$

and the MAP estimate

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- closed-form solution

$$\hat{w}_{\text{MAP}} = (\Phi(x)^\top \Phi(x) + \lambda I_{M \times M})^{-1} \Phi(x)^\top y, \quad \lambda = \frac{\sigma^2}{\tau^2}$$

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- For Lasso (L1):
 - Prior: Laplace distribution $p_W(w) \propto \exp(-\lambda_l \|w\|_1)$.

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- No closed-form solution! But we can use convex optimization algorithms to solve it!

Ridge Regression: Closed-Form Solution

- L2-regularized objective:

$$J(w) = \frac{1}{2} \|\mathbf{y} - \Phi(\mathbf{x})w\|^2 + \frac{\lambda}{2} \|w\|^2$$

Ridge Regression: Closed-Form Solution

- L2-regularized objective:

$$J(w) = \frac{1}{2} \|\mathbf{y} - \Phi(\mathbf{x})\mathbf{w}\|^2 + \frac{\lambda}{2} \|\mathbf{w}\|^2$$

- Closed-form solution (similar to LS):

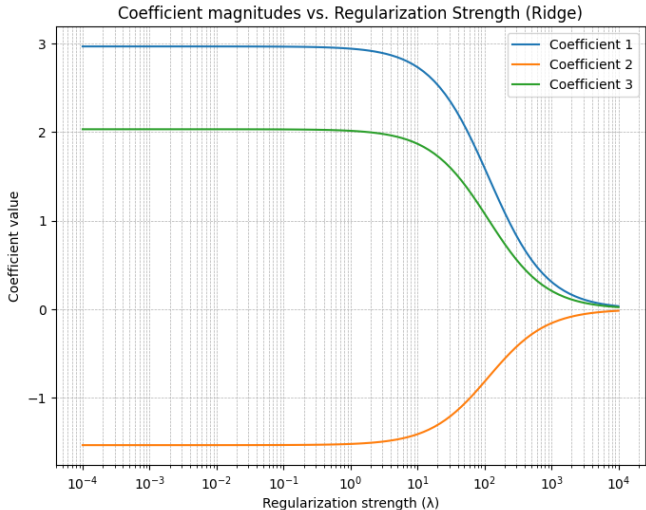
$$\hat{\mathbf{w}}_{\text{ridge}} = (\Phi(\mathbf{x})^\top \Phi(\mathbf{x}) + \lambda \mathbf{I}_{M \times M})^{-1} \Phi(\mathbf{x})^\top \mathbf{y}$$

- Always invertible ($\lambda > 0$ ensures positive definite).
- Numerically more stable.

Ridge Regression: Closed-Form Solution

- L2-regularized objective:

$$J(w) = \frac{1}{2} \|y - \Phi(x)w\|^2 + \frac{\lambda}{2} \|w\|^2$$



L1 Regularization (Lasso)

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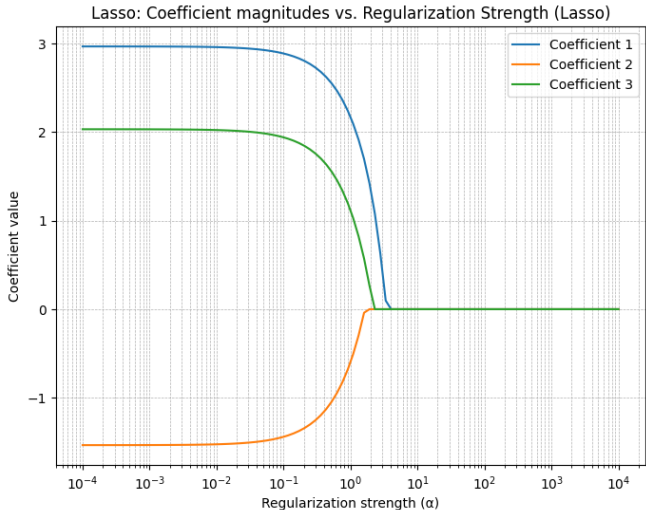
$$J(w) = \frac{1}{2} \|\mathbf{y} - \Phi(\mathbf{x})w\|^2 + \lambda \|\mathbf{w}\|_1$$

- Key properties:
 - Induces *sparsity* (exactly zero coefficients).
 - Performs feature selection automatically (since it can choose exactly zero coefficients).
 - No closed-form solution (requires convex optimization).

L1 Regularization (Lasso)

- L1-regularized objective:

$$J(w) = \frac{1}{2} \|y - \Phi(x)w\|^2 + \lambda \|w\|_1$$



L1 vs L2 Regularization

Property	L2 (Ridge)	L1 (Lasso)
Solution type	Closed-form	Iterative optimization
Sparsity	No	Yes
Feature selection	No	Yes
Robustness to outliers	Less	More
Prior distribution	Gaussian	Laplace

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- Elastic Net: Combines L1 + L2

$$R(w) = \alpha \|w\|_1 + (1 - \alpha) \frac{1}{2} \|w\|_2^2$$

Balances feature selection (L1) and stability (L2)

Normalization

- To avoid numerical problems (e.g., a large conditioning number on $\Phi(\mathbf{x})$), it is standard to normalize the matrix $\Phi(\mathbf{x})$ by applying:
 - **Centering:** Subtract the mean from each column (feature) of $\Phi(\mathbf{x})$ so that each has zero mean.
 - **Scaling:** Divide each centered feature by its standard deviation so that each has unit variance.
- In mathematical terms, for the j -th feature:

$$\tilde{\phi}_j(x_n) = \frac{\phi_j(x_n) - \mu_j}{\sigma_j}, \quad \text{with } \mu_j = \frac{1}{N} \sum_{n=1}^N \phi_j(x_n) \text{ and } \sigma_j = \sqrt{\frac{1}{N-1} \sum_{n=1}^N (\phi_j(x_n) - \mu_j)^2}.$$

Summary

- Regularization helps control model complexity
- Different regularizers impose different priors:
 - L2: Gaussian prior $\rightarrow$ small weights
 - L1: Laplace prior $\rightarrow$ sparse weights
- Practical considerations:
 - Use L2 for correlated features, and numerical stability.
 - Use L1 for feature selection, and high-dimensional data.
 - Cross-validate λ for optimal performance